NAME: _____ Geometry (FL B.E.S.T.)

12.3 Conditional Probability

Conditional probability asks: given that event B has already happened, how likely is A? The formula is $P(A \mid B) = \dfrac{P(A \text{ and } B)}{P(B)}$ — restrict attention to the world where B is true, then ask what fraction of *that* world also has A. Two-way (contingency) tables make this concrete: read down a column (or across a row) to get the reduced sample space. A key fact worth remembering: $P(A \mid B)$ and $P(B \mid A)$ are generally **not equal** — order matters. When $P(A \mid B) = P(A)$, knowing B told you nothing new about A, and the events are independent (Lesson 12.2).

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. $P(A \text{ and } B) = 0.15$, $P(B) = 0.6$. Find $P(A \mid B)$.

2. Find $P(\text{ace} \mid \text{red card})$.

3. Find $P(6 \mid \text{even})$.

4. $P(A) = 0.5$, $P(A \mid B) = 0.4$. Are they independent?

5. 50 people; 20 take the bus; 12 of the bus riders are students. Find $P(\text{student} \mid \text{bus})$.

6. Find $P(\text{odd} \mid \text{less than 4})$.

7. $P(A \text{ and } B) = \dfrac{1}{10}$, $P(A) = \dfrac{2}{5}$. Find $P(B \mid A)$.

8. Find $P(\text{heart} \mid \text{red})$.

9. Events are independent exactly when $P(A \mid B)$ equals what?

10. Bag: 5 red, 5 blue. Find $P(\text{red} \mid \text{not green})$.

11. 80 voters; 48 said yes; 30 of the yes voters are under 30. Find $P(\text{under 30} \mid \text{yes})$.

12. Is $P(A \mid B)$ always the same as $P(B \mid A)$?

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13. $P(A \text{ and } B) = 0.12$, $P(A) = 0.3$. Find $P(B \mid A)$.

14. Of 120 students, 72 take art; 45 of the art students also take music. Find $P(\text{music} \mid \text{art})$.

15. A rare condition affects 2% of a population. A test is 95% accurate both ways. Using a population of 10,000, find $P(\text{condition} \mid \text{positive test})$.

✓ Answer Key

1. 0.25
2. $\frac{1}{13}$
3. $\frac{1}{3}$
4. No
5. $\frac{3}{5}$
6. $\frac{2}{3}$
7. $\frac{1}{4}$
8. $\frac{1}{2}$
9. $P(A)$
10. $\frac{1}{2}$
11. $\frac{5}{8}$
12. No, not in general
13. 0.4
14. $\frac{5}{8}$
15. $\frac{19}{68}$